

WorldDeciding Installation Instructions

Prerequisites

- .NET 8 SDK for building and running the ASP.NET Core API.
- Node.js and npm for installing and building the React/Vite frontend.
- Docker Desktop or Docker Engine with Docker Compose for PostgreSQL, Redis, and containerized services.
- A Gemini API key if AI summary features will be used.

Repository Layout

- WorldDeciding: ASP.NET Core API project.
- WorldDeciding.Application: commands, queries, DTOs, validators, and interfaces.- WorldDeciding.Domain: entities and identity models.
- WorldDeciding.Infrastructure: EF Core, Redis, email, GeoIP, token, and AI implementations.
- worlddeciding-frontend: React, TypeScript, Vite, and Tailwind frontend.- documentation: generated thesis, presentations, and support documents.

Environment Variables

- Create or update .env before Docker execution.
- Set POSTGRES_PASSWORD, JWT_KEY, REFRESHTOKEN_PEPPER, PRIVACY_IPSALT, GEMINI_API_KEY, and PGADMIN_PASSWORD.
- Optionally set POSTGRES_USER, POSTGRES_DB, FRONTEND_BASE_URL, API_BASE_URL, CORS_ALLOWED_ORIGIN_0, CORS_ALLOWED_ORIGIN_1, GEMINI_MODEL, and GEMINI_API_VERSION.

Docker Compose Run

- From the repository root, run: docker compose up -d --build
- The API container listens on port 8080 inside the compose configuration.
- PostgreSQL is published on host port 5433, Redis on 6379, pgAdmin on 5050, and RedisInsight on 5540.

- Check container health with: `docker compose ps`.
- Stop the stack with: `docker compose down`.

Backend Development Run

- Restore packages with: `dotnet restore WorldDeciding.sln`
- Build with: `dotnet build WorldDeciding.sln`
- Apply EF Core migrations to the configured PostgreSQL database when required.
- Run the API with: `dotnet run --project WorldDeciding/WorldDeciding.csproj`
- Use `appsettings.Development.json` and local environment variables for development settings.

Frontend Development Run

- Open `worlddeciding-frontend`.
- Install packages with: `npm install`
- Run the development server with: `npm run dev`
- Build production assets with: `npm run build`
- Configure the frontend API base URL according to the environment used by the API.

Post-Install Verification

- Open the frontend and confirm that public pages load.
- Register a test account, confirm email behavior according to the configured email provider, and login.
- Create or publish a question as an admin user, vote as a registered user, and open statistics.
- Verify Redis-backed rate limits and AI summary behavior only after required secrets are configured.